AMALITECH DATA ANALYTICS & ENGINEERING CAPSTONE

MarketPulse Ghana

A Modern Analytics Engineering Platform for Food Commodity Price Intelligence

39,038
Verified Price Records

17 Years
211 Continuous Months (2006–2023)

26 Staples
Cereals, Roots, Tubers, Pulses

44 Markets
10 Legacy Administrative Regions

28 Tests
Automated CI/CD Contracts Passing

Developer: Capstone Data Group 2

Live Platform: marketpulse-ghana.vercel.app

PROBLEM & MOTIVATION

Why Ghana's Food Commodity Markets Needed Modern Analytics

1. Extreme Price Volatility

- In late 2022, headline food inflation in Ghana surged past 54%, devastating households and businesses.
- Rapid price spikes created severe uncertainty for agricultural aggregators and processors.
- Decision-makers struggled to separate local crop shortages from national monetary shocks.

2. Confusing Packaging Units

- Food commodities are sold in non-standard units: 100kg bags, 2.5kg olonkas, 100–tuber bundles, crates, and sacks.
- Comparing a 100–tuber yam bundle to a 50kg bag of rice without standardization produces meaningless numbers.
- Raw price series exhibit wild zig-zags from inconsistent unit definitions.

3. Locked & Fragmented Data

- Over 17 years of high-quality WFP survey records remained trapped in static government CSV files.
- Zero interactive filtering, no automated unit conversion, and no unified historical dashboard.
- Field data lacked automated quality gates, leaving errors undetected.

4. Severe Information Opacity

- Smallholder farmers lack visibility into Accra wholesale prices, losing bargaining power to middlemen.
- Agribusiness aggregators struggle to time bulk procurement around seasonal harvest gluts.
- Policymakers lack continuous per-kg benchmarks across Ghana's 10 regions.

CORE MISSION: **Standardize all packaging into Physical Mass Benchmarks (GHS/kg), fix structural survey errors, and deliver trusted, interactive intelligence at zero ongoing database infrastructure cost.**

DATASET SCOPE & PROVENANCE

17 Years of UN WFP & MoFA Market Price Surveys

Dataset Specifications & Provenance

- **Data Source:** United Nations World Food Programme (WFP) & Ministry of Food and Agriculture (MoFA) via HDX.
- **39,038 Price Records** spanning 211 consecutive months from January 2006 to July 2023.
- **26 Monitored Food Staples:** Maize, Rice, Yam, Cassava, Plantains, Sorghum, Millet, Cowpeas, Groundnuts, Fish, Meat.
- **44 Physical Market Centers:** From rural surplus production hubs (Techiman, Tamale, Ejura) to major urban consumption centers (Accra, Kumasi, Takoradi).
- **Spatial Provenance (10 vs 16 Regions):** Ghana reorganized into 16 regions in 2018. We standardize all data to the 10 legacy regions to preserve unbroken 17-year spatial continuity.
- **Value Chain Tiers:** Wholesale commercial bulk transactions and Retail consumer checkout prices.

Key Spatial Decision: **Splitting regions mid-dataset in 2018 would have broken time-series comparisons. Mapping modern district points back to 10 legacy regions ensures 100% longitudinal validity.**

STRATEGIC ARCHITECTURE PIVOT

From Premature ML to Rock-Solid Analytics Engineering

1. Initial Machine Learning Focus

- **Original Objective:** Train XGBoost regression models to forecast 1-month-ahead wholesale maize prices.
- **Why Wholesale Maize?** Selected because it offers 209 months of continuous coverage (longest unbroken series). Retail maize had only 48 months.
- Engineered 18 temporal lag features, rolling statistics, and 9-fold expanding walk-forward cross-validation.

2. The Reality Check during EDA

- Exploratory profiling uncovered fundamental survey flaws: 21.6% of records had retail prices lower than wholesale.
- Naive SQL LAG() produced 46 false MoM and 185 false YoY inflation spikes across survey gaps.
- Packaging unit mismatches (bags vs bowls) meant models were learning data hygiene noise, not economic signal.
- Stakeholders couldn't trust historical numbers, making predictive modeling premature.

3. The Strategic Pivot & Asset Preservation

- Pivoted engineering effort to build a Modern Analytics Stack: staging views, dimensional models, marts, and testing.
- **Preserved All ML Assets:** Feature pipelines, 9-fold CV results, and tuned model weights archived in **archive/ml/**.
- **Stress-Tested Holdout:** Evaluated model on Jan 2022 – Jun 2023 (peak currency volatility crisis).
- Authored **INFERENCE_PROTOCOL.md** defining exact downstream operational schema.

ENGINEERING LESSON: **Predictive forecasting on unstandardized, flawed data produces misleading predictions. Solid analytics engineering foundations must come first.**

MODERN DATA STACK ARCHITECTURE

High-Performance, Zero-Cost Cloud Analytics Engine

1. Ingest & Telemetry

- **scripts/ingest_hdx.py** streams monthly CSV updates directly from WFP HDX.
- Exponential-backoff retry (3 attempts) & 50-row minimum integrity floor.
- Snapshots **raw_wfp_prices_previous** before table recreation for change audit.
- Writes execution metadata to **meta_pipeline_runs** with UTC TIMESTAMPTZ.

2. Staging & Dimensions

- **stg_food_prices**: per-kg price normalization, canonical **unit_family** classification.
- **stg_markets**: coordinate casting, Title Case administrative region normalization.
- **dim_commodities**: 26 crop catalog.
- **dim_markets**: 44 nodes + **market_tier**.
- **dim_calendar**: 211 months with **agricultural_season** domain encoding.

3. Marts & Governance

- **mart_commodity_monthly**: strict DATEDIFF calendar MoM/YoY and per-kg aggregates.
- **mart_regional_spreads**: per-kg spatial spread % vs national median.
- **mart_wholesale_retail_margins**: verified commercial markups aggregated by MEDIAN.
- **mart_pipeline_health**: observability view.
- **audit_data_anomalies**: quarantined records.

4. Testing & Delivery

- **scripts/test_marts.py** executes 28 automated quality contract assertions.
- Zero-infrastructure runtime: DuckDB builds all marts in ~1 second.
- GitHub Actions runs monthly data sync.
- Automated Pull Request with live Vercel Preview deployment for human review.
- Evidence.dev compiles to static WASM/Parquet for global sub-second queries.

Python 3.12 | DuckDB 1.0+ Columnar Engine | Evidence.dev Front-End | DuckDB-WASM | Vercel Edge ($0 Hosting Cost) | GitHub Actions CI/CD

DATA HYGIENE & INTEGRITY

Resolving Critical Traps in Agricultural Survey Data

1. Temporal Gaps in MoM & YoY Calculations

- **Trap:** Standard SQL LAG() by row sequence compares records across skipped months or multi-year survey breaks, producing 46 false MoM spikes and 185 false YoY spikes.
- **Fix:** Enforced strict calendar DATEDIFF intervals (27–35 days for MoM, 350–380 days for YoY). Non-contiguous observations safely return NULL.

2. Canonical Unit Family Classification

- **Trap:** Commodities were recorded in a chaotic mix of mass units (KG, Bag) and piece counts (Tubers, Olonkas), corrupting volume comparisons.
- **Fix:** Standardized every unit into exactly two canonical labels: 'Mass / Weight' and 'Piece / Count'. Validated permanently by an automated assertion rule.

3. Multi-Packaging Unit Zigzags

- **Trap:** Mixing 50kg bags, 100kg bags, and retail bowls on the same trend line created artificial price volatility on analytical charts.
- **Fix:** Staple time-series are filtered strictly to commercial standard packaging (e.g. 100kg for Maize), with clean per-kg median calculations.

4. Margin Inversions & MEDIAN Aggregation

- **Trap:** 21.6% of WFP wholesale-retail pairs had retail prices lower than wholesale due to mismatched field enumerator measurement methods.
- **Fix:** Flagged **is_verified_margin** (retail >= wholesale and markup <= 200%). Quarantined 3,090+ records to **audit_data_anomalies**; aggregated verified markups via MEDIAN.

RESULT: 3,090+ flawed records quarantined with 100% transparency. Historical metrics are verified, clean, and outlier-resistant.

DOMAIN MODELING IN SQL

Agricultural Seasons & Market Commercial Tiers

1. dim_calendar.agricultural_season

- Encodes Ghana's agro-ecological cycle into every calendar month across 17 years:
- **Lean Season (May–Jul):** Northern household granaries run dry ahead of new harvest; annual price peaks for cereals and pulses across northern transit markets.
- **Major Harvest (Aug–Nov):** Southern bimodal & northern unimodal harvest glut; seasonal supply flood brings prices to annual floor.
- **Post-Harvest (Dec–Jan):** Marketing, commercial bagging, and aggregation period.
- **Minor Season (Feb–Apr):** Southern minor rains and secondary crop harvest.
- **Impact:** Analysts and SQL queries filter by harvest phase directly without requiring external agro-ecological lookup tables.

2. dim_markets.market_tier

- Establishes a 3–tier commercial hierarchy for all 44 monitored market centers:
- **Major National Hub:** High-volume coastal consumption centers (Accra, Tema) and central transit hubs (Kumasi Central).
- **Regional Center:** Strategic regional capital markets and assembly hubs (Tamale, Sunyani, Cape Coast, Bolgatanga, Wa, Ho, Koforidua).
- **Local Market:** Rural assembly points, farming district markets, and border transit nodes.
- **Impact:** Enables systematic spatial comparison between urban consumption centers and rural production zones to measure market integration.

WHY THIS MATTERS: Real-world domain knowledge is baked directly into the dimensional schema, allowing any analyst or dashboard user to ask agro-economic questions natively in SQL.

KEY FINDING 1: MACROECONOMICS

Currency Shocks vs. Real Agricultural Scarcity

Historical Price Trajectories (2006–2023)

- **2006 to 2021:** Food prices followed moderate secular inflation (5%–8% annually), fluctuating predictably according to harvest cycles.
- **2022 to 2023 Price Surge:** Prices exploded by over 65% Year-over-Year across all 26 monitored food staples.
- **Nominal vs. Real Pricing:** Price surveys are reported in Nominal Ghanaian Cedis (GHS). The 2022 surge was driven by severe national currency depreciation and imported diesel costs — not an agricultural crop failure.
- **Policy Implication:** A currency shock requires macroeconomic monetary intervention, whereas crop failure requires emergency food reserves. MarketPulse Ghana's nominal series is transparent about this limitation in the governance page.

KEY FINDING 2: SPATIAL DISPARITIES

Transit Friction & Per-KG Regional Price Wedges

Spatial Arbitrage & Commodity Perishability

- **Bulky Perishables Have Highest Regional Wedges:** Yam and Cassava show markups up to +40% in coastal consumer markets (Accra) compared to farming hubs in Brong-Ahafo and Northern regions.
- **Transit Costs Dictate Spread:** Root tubers are heavy relative to value and spoil quickly. Road infrastructure gaps and diesel price inflation widen these spreads immediately.
- **Grains Are Well Integrated:** Dry cereals (Maize, Millet, Sorghum) show narrow 10%–15% regional spreads because they store well and traders actively arbitrage price differences across regions.
- **Quantified in mart_regional_spreads:** Computes native per-kg spread % against national median for every commodity, market, and time period.

KEY FINDING 3: SEASONALITY & PROCUREMENT

Capitalizing on Harvest Gluts to Reduce Food Costs

Agro-Ecological Price Cycles & Cost Savings

- **Bimodal South vs. Unimodal North:** Southern Ghana benefits from major (Aug–Sep) and minor (Nov–Dec) harvests. The North relies on a single harvest (Oct–Nov).
- **The Lean Season Price Spike:** Every year between May and July (Lean Season), grain prices reach their annual peak across northern markets as household granaries run dry.
- **Commercial Value:** Agribusiness aggregators, poultry feed mills, and food processors can save 20% to 35% on raw material costs by timing bulk procurement to the September–October Major Harvest glut.
- **Direct SQL Access:** `dim_calendar.agricultural_season` provides an instant handle for commercial procurement analytics.

PRODUCTION TESTING & GOVERNANCE

Automated Quality Gates, Observability & CI/CD

1. 28-Check Assertion Suite

- **scripts/test_marts.py** executes 28 automated tests before any deployment:
- **Table/Mart Existence:** 10 tests verifying all views and tables are generated.
- **Primary Key Uniqueness:** 6 tests preventing duplicate dimensional records.
- **Non-Null Mandatory Fields:** 5 tests checking essential schema columns.
- **Domain Logic & Value Sanity:** 7 tests enforcing price > 0, margin validity, canonical **unit_family** labels, and temporal span.
- **Status:** 28/28 Passing (100%), Hard exit-code 1 gate on failure.

2. Observability & Transparency

- **audit_data_anomalies**: Quarantines 3,090+ inverted margins and extreme price outliers for full auditability — zero silent drops.
- **meta_pipeline_runs**: Logs UTC execution start, duration, record count, and exit status for every pipeline step.
- **mart_pipeline_health**: Dedicated Evidence.dev-facing view exposing real-time operational status on the governance dashboard.
- Zero-cloud cost infrastructure: Columnar transformations run in DuckDB without recurring database server fees.

3. Monthly CI/CD with PR Review

- Automated GitHub Actions workflow runs on the 1st of every month at 06:00 UTC.
- Executes ingestion -> rebuilds marts -> runs 28 test assertions -> updates UI DuckDB sources.
- **Enterprise Governance Gate:** Pipeline does not push directly to production.
- Opens an automated Pull Request with a live Vercel Preview deployment for human sign-off before merging.

TEST SUITE: 28/28 Data Contracts Passing • Zero Pipeline Failures • Vercel Preview Human Sign-Off Gate

FUTURE HORIZON: PREDICTIVE ANALYTICS

Downstream Machine Learning Inference Protocol

1. Preserved ML Assets

- All Phase 2 (features) and Phase 3 (modeling) notebooks preserved in **archive/ml/**.
- Serialized XGBoost model: **archive/ml/models/xgboost_maize_wholesale.json**.
- 9-fold walk-forward cross-validation outputs documented with full feature weights.

2. Stress-Tested Holdout

- Evaluated on January 2022 – June 2023 holdout period.
- Deliberate stress test during peak currency crisis and inflation spike (>65% YoY).
- Proves model behavior under the most volatile out-of-sample economic conditions in Ghana's history.

3. Inference Protocol

- Formal contract in **archive/ml/INFERENCE_PROTOCOL.md**.
- Reads clean, validated monthly actuals from **mart_commodity_monthly**.
- Generates 1-to-3 month price projections with 95% confidence intervals.
- Writes output to **mart_maize_forecasts** for UI display.

4. Retraining Cadence

- Bi-annual retraining aligned with August (Major Harvest) and November (Minor Harvest) syncs.
- Automated drift triggers: Retrain if rolling 3-month MAPE exceeds 15% or inflation exceeds 40%.
- Seamless integration onto new **ui/pages/forecasts.md** Evidence page.

INTEGRATION READY: The analytics foundation is complete. When stakeholders request predictive forecasts, the inference pipeline plugs directly into trusted marts.

CAPSTONE DEFENSE CONCLUSION

Delivering Transparent Market Intelligence for Ghana

Core Project Achievements

- **End-to-End Analytics Platform:** Automated ingestion, staging, dimensional modeling, and 4 analytical marts.
- **39,038 Price Records Normalized:** 17 continuous years standardized to physical mass benchmarks (GHS/kg).
- **Domain Knowledge Baked In:** `agricultural_season` and `market_tier` encode Ghana's food system directly into SQL.
- **Enterprise Quality Standards:** 28-rule automated test suite passing in CI/CD with 0 failures.
- **Zero-Infrastructure Web UI:** Serverless Evidence.dev dashboard with DuckDB-WASM browser queries deployed on Vercel.
- **Complete Operational Governance:** Anomaly quarantine view, pipeline audit logging, and PR-review deployment gates.

Live Verification & Codebase

- **Automated Pipeline:** Monthly GitHub Actions cron with PR preview

Live Application: marketpulse-ghana.vercel.app

GitHub: github.com/primegideon/marketpulse-ghana